

BUSS386 Problem Set 2

Interest Rates

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Problem 1 — Compounding conversion

A bank quotes you an interest rate of 14% per annum with quarterly compounding.

- (a) Find the equivalent rate with continuous compounding.
- (b) Find the equivalent rate with annual compounding.
- (c) Without computing, predict whether (a) is higher or lower than (b).

Problem 2 — Zero rates and bond pricing

The 6-month, 12-month, 18-month, and 24-month zero rates (continuously compounded) are:

Maturity	0.5y	1y	1.5y	2y
Zero rate (% c.c.)	4.0	4.5	4.75	5.0

- (a) Find the price of a 2-year bond with face value \$100 that pays a 5% per-annum coupon *semiannually* (so \$2.50 every 6 months, plus \$100 face at maturity).
- (b) Compute the 2-year *par yield* (continuously compounded) — the coupon rate that makes the bond price equal to its face value.
- (c) Explain why the par yield in (b) is close to (but not exactly equal to) the 2-year zero rate.

Problem 3 — Forward rates and FRAs

Use the same zero-rate term structure as Problem 2.

- (a) Compute the forward rate (continuously compounded) for the period from 18 months to 24 months, $r_0(1.5, 2)$.
- (b) Compute the forward rate for the period from 6 months to 12 months, $r_0(0.5, 1)$.
- (c) Convert the forward rate from (a) to its semiannually compounded equivalent.
- (d) An FRA promises to pay you 7% (semiannually compounded) and receive 6-month SOFR on a notional of \$10,000,000 for the 6-month period starting 18 months from now. Compute the value of the FRA today (use the 24-month zero rate to discount).
- (e) What does the sign of your answer in (d) tell you about whether you would prefer to receive fixed or pay fixed at the current forward rate?

Problem 4 — Bond duration

A 5-year bond pays an 8% annual coupon (one payment per year) and has a yield-to-maturity of 11% per annum (continuously compounded). Face value \$100.

- (a) Compute the bond's price.
- (b) Compute the bond's Macaulay duration (which equals modified duration here, since the yield is continuously compounded).
- (c) Use duration to estimate the new price if the yield falls by 0.20 percentage points (from 11.00% to 10.80%).
- (d) Recompute the price at $y = 10.80\%$ exactly, and compare with your duration-based estimate from (c). How close are they?

Problem 5 — Duration-matched portfolios and convexity

Two portfolios, both yielding 10% (continuously compounded):

- **Portfolio A (barbell):** a 1-year zero with face \$2,000 plus a 10-year zero with face \$6,000.
 - **Portfolio B (bullet):** a 5.95-year zero with face \$5,000.
- (a) Compute the value and Macaulay duration of each portfolio. Verify that they have (essentially) the same duration.
 - (b) For a small parallel yield rise of 0.10% (from 10.00% to 10.10%), use duration to estimate the percentage change in each portfolio's value. Are they (essentially) equal?
 - (c) For a *large* parallel yield rise of 5% (from 10% to 15%), recompute each portfolio's value *exactly*. Compute the actual percentage change in each. Which portfolio loses less?
 - (d) Explain the result in (c) using the concept of *convexity*.